Tanvi Randhir Badghare

Graduate Data & Performance Engineer | Motorsport Telemetry, Python, Statistics | Open to relocation (Italy, UK, EU) |
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Profile

Computer Science & Engineering graduate (CGPA 8.53/10) with hands-on experience in motorsport telemetry analysis, statistical comparisons, and data visualisation. Built tools for lap-by-lap driver comparison, telemetry processing, and secure data pipelines, with an emphasis on reproducible analysis and evidence-based conclusions. ery eager to learn tyres and vehicle dynamics from race engineers.

Education

VIT Bhopal University Bhopal, Madhya Pradesh, India

B.Tech. Computer Science & Engineering (Cyber Security and Digital Forensics) Oct. 2022 – Oct. 2026

CGPA: 8.53/10 | IELTS: 7.5 overall | Relevant coursework: Computer Networks, Operating Systems, Data Structures and Algorithms, Applied Cryptography, Database Management Systems, Object-Oriented Programming.

Engineering Projects

F1 Performance Analyser 2026

Python, OpenF1, Pandas

- Built a two-driver Grand Prix analyser; on real Japan 2026 data it measured a 0.212 s median pace gap (Piastri vs Norris, 41 paired laps, 95% CI +0.093 to +0.349 s, 71% of laps won), shrinking to 0.093 s in clear air.
- Validated distance-aligned telemetry against official sector times (error 0.24 s to under 0.01 s) and repaired stale source-data samples.
- Added guards against unsupported tyre-degradation and setup claims; 36 tests plus an OpenF1 loader tested on a simulated server.

PitCrypt-F1: Secure Telemetry Pipeline Mar. 2026 – Oct. 2026

Python, Pandas, Streamlit, Plotly

- Built a Car–Relay–Validator pipeline processing F1 telemetry, including speed, RPM, throttle, braking, gear, DRS, and track position, with an interactive dashboard.
- Implemented packet validation, signatures, authenticated encryption, replay protection, anomaly filtering, and structured audit logging.
- Benchmarked simulated 10-car workloads at 9.35k packets/s aggregate throughput and 0.71 ms median run-mean latency.

SecureCAN: Vehicle Network Simulator Aug. 2026 – Present

C++17, Vehicle Telemetry

- Developed a deterministic CAN-network simulator modelling three ECUs using shared vehicle state and synthetic or telemetry-derived inputs.
- Implemented physics-based plausibility detection and measured detection latency and resulting vehicle-state error.
- Validated with 353 automated tests, sanitizer-clean builds, and reproducible runs.

RaceIntel: Trust or Bluff 2026

React Native, JavaScript

- Built an interactive F1 strategy game demonstrating how spoofed or corrupted telemetry can mislead race decisions.

Distributed Web Application Firewall (D-WAF) — Capstone | Technical Lead (5) Jan. 2025 – Jan. 2026

Rust, Go, Distributed Systems

- Led a five-member team developing a distributed WAF with ML-based threat detection; implemented core components and coordinated integration with a teammate's STIX/TAXII module.

Technical Skills

Data Analysis: Python, Pandas, NumPy, Paired Comparisons, Confidence Intervals, Statistical Analysis

Visualisation & Applications: Plotly, Streamlit, Matplotlib

Motorsport Data: OpenF1, FastF1, Lap and Sector Analysis, Distance-aligned Telemetry, Stint Comparisons

Systems & Tools: C++17, CAN, Git, GitHub, Linux